

ENGR 2303 Electrical Science

Course Catalog Description: This course teaches analysis techniques for electrical circuits which consist of resistors, capacitors, and inductors. The circuits analyzed are driven by constant and sinusoidal voltage and current sources.

Course Prerequisites: PHY 2114 with a minimum grade of “C” and ENGR 2311 or concurrent enrollment in ENGR 2311.

Course Co-requisites: ENGR 2303D.

Prerequisites by Topic: 1. Ohm's Law 2. Basic mathematical operations with complex number

Textbook: “Fundamentals of Electric Circuits”, 2026 Release by Alexander and Sadiku. ISBN: 978-1-266-02040-7

Resources:

Course Website: <https://EngineeringWithBingabr.github.io/>

Video Lectures: <https://www.youtube.com/@EngineeringWithBingabr>

Instructor: Mohamed Bingabr, Professor of Engineering.

Office Location: Howell Hall 221C

Phone: (405) 974-5718

Email: mbingabr@uco.edu

Course Meeting Time: TR 8:00 am – 9:15 am

Course Meeting Location: Howell Hall 205

Office Hours: TR 3:00 to 5:25 pm, and by appointments.

Overall Education Objective: To introduce methods and techniques for electrical circuit analysis and design with DC and AC sources.

Course Learning Objective: At the end of this course, students should be able to:

1. Understand and analyze dc and ac electrical circuits with resistors, capacitors, inductors, operational amplifier, independent and dependent sources;
2. Understand and use ohm's and Kirchhoff's laws in circuit analysis;
3. Understand and use Nodal and Mesh analysis in circuit analysis;
4. Understand and use superposition, Thevenin's, and Norton's theorems in circuit analysis;
5. Analyze and design circuits with operational amplifiers;
6. Conduct ac and dc power analysis of electrical circuits;
7. Conduct frequency analysis of electrical circuits.

Topics Covered:

Subject	Reading
Basic Concepts and Resistive Circuits Ohm's Law, Kirchhoff's Law Series and Parallel Resistor	Ch1 Ch2
Nodal Analysis, Loop Analysis Test 1	Ch3
Additional Analysis Techniques Superposition, Thevenin's and Norton's Theorem Maximum Power Transfer Test 2	Ch4
Operational Amplifiers, Op-Amp Models Fundamental Op-Amp Circuits, Comparators	Ch5
Capacitance and Inductance Capacitor and Inductor Combinations	Ch6

RC Operational Amplifier Circuits	
First Order Circuits Test 3	Ch7
AC Steady-State Analysis Sinusoidal and Complex Forcing Functions Phasors, Impedance and Admittance	Ch9 Ch 10
Steady-State Power Analysis Instantaneous and Average Power Maximum Average Power Transfer Effective or rms Values	Ch11 11.1-11.3
Variable-Frequency Network Performance Variable-Frequency Response Analysis Sinusoidal Frequency Analysis Resonant Circuits, Filter Networks Test 4	Ch14 14.7

Distribution of Points:

Homework 10% Attendance 10% Quizzes 10% 4 Tests 70%

Grading Scale: A: 90-100; B: 80-89; C: 70-79; D: 60-69; F: 0-59

COURSE EXPECTATION & CONDUCT:

Attendance is expected. You will be responsible for any announcements or notes from class. It is expected that each student will spend a total of 7 hours per week on the course (not including lecture times). I don't expect you to memorize formulas, but I expect you to understand them. So, you are allowed to bring the formula sheets posted on the course website and one page of your own notes to the quizzes and exams. Should you miss a quiz or test due to illness or an emergency, you will be required to give advance notice or provide a doctor's excuse to be allowed to make-up on the tests and quizzes you missed. Make-up tests and quizzes are usually harder than the regular tests and quizzes given during the class. Cheating or academic dishonesty of any kind will not be tolerated. NO CELL PHONE DURING LECTURES.

HOMEWORK:

The reality is this: you will not truly understand the technical concepts by just paying attention to lectures and reading the materials and examples in the textbook. To truly understand engineering concepts, you must solve the homework problems yourself even if you are struggling with solving them. For this reason, regular homework assignments will be made. The quizzes will be like homework problems. Homework will be due at the beginning of the class period on the due date. Late homework will be penalized accordingly (25% reduction each day). Homework will be submitted in D2L.

ADA STATEMENT:

"The University of Central Oklahoma complies with Section 504 of the Rehabilitation Act of 1973 and the American with Disabilities Act of 1990. Students with disabilities who need special accommodations should make their requests by contacting the coordinator of Disability Support Services, Kimberly Fields at 974-2549/2516, DSS@uco.edu. The office is in the Nigh University Center, Room 415. Students should also notify the instructor of special accommodation needs by the end of the first week of class."

STUDENT INFORMATION SHEET

<https://www.uco.edu/academic-affairs/files/student-info-sheet.pdf>

Prepared by:	Mohamed Bingabr, Associate Professor of Engineering and Physics
Date:	August 16, 2026

